INA236

User manual

DC current and voltage measurement module

ESP32 · Arduino IDE · I2C

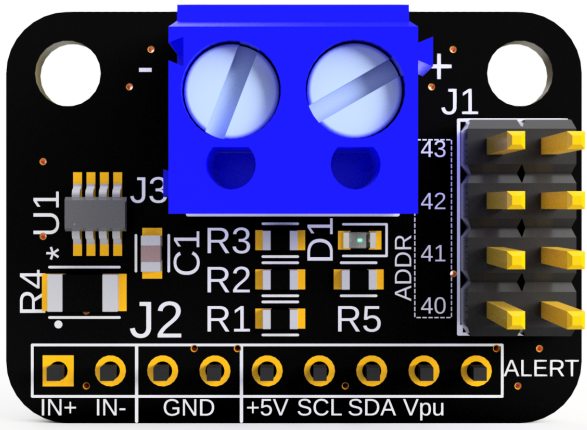


Figure 1. Front view of the module. Image supplied by the designer.

Two connection paths, two calibration values.
Input A: lower header, 25 mΩ, 0 to 3.2 A.
Input B: blue terminal block, 31 mΩ equivalent, calculated 0 to 2.58 A.
Measurement bus voltage: 0 to 48 V DC.

Before connecting

The pin marked +5V accepts a 3.3 V or 5 V supply. Vpu sets the I2C pull-up voltage: use 3.3 V with ESP32. The 48 V rating applies to the measured circuit, never to the supply or digital pins.

Version 1.0 · October 16, 2026 · English

IMPORTANT BEFORE CONNECTING
Blue terminal block: the physical board has no + or - symbols on its silkscreen. In the orientation of Figure 1: **right = IN+; left = IN-**. Symbols in the image are visual references only.
Common ground is required: connect measured-circuit GND, INA236 GND and ESP32 GND together. Without this shared reference, voltage will not be measured correctly. See page 2 for details.

01 / Connections and wiring

9-pin J2 header: manual reference numbering, left to right when viewing the module as shown in Figure 1.

Pin	Terminal	Function / connection
1	IN+	Positive measurement input A; toward positive source.
2	IN-	Negative measurement input A; toward load positive.
3	GND	Common ground for supply, microcontroller and measured circuit.
4	GND	Second common ground connection.
5	+5V	Module supply: 3.3 V or 5 V.
6	SCL	I2C clock. ESP32: GPIO22.
7	SDA	I2C data. ESP32: GPIO21.
8	Vpu	I2C pull-up supply: 3.3 V for ESP32; 5 V only for compatible logic.
9	ALERT	Alert output. ESP32: GPIO27; external 4.7 kΩ pull-up to 3.3 V.

Current path: select A or B

Source (+) → IN+ → module → IN- → load (+).
Source (-) → load (-) and module/ESP32 GND.

For A, use pins 1 and 2. For B, use the blue terminal block: in Figure 1 the right terminal is (+)/IN+ and the left terminal is (-)/IN-. These are alternative paths through the same sensor, not two independent channels. Use one path at a time.

Wire with all supplies off. IN- goes toward the load and is not the GND return. Do not connect IN+ and IN- directly across the source positive and negative terminals: the shunt has very low resistance. The module provides no galvanic isolation.

NOTICE: THE TERMINAL BLOCK HAS NO + / - MARKINGS
The + and - symbols shown in Figure 1 **are not printed on the physical board silkscreen**. The orientation stated in this manual is correct: view the module from the front as in Figure 1, with the blue terminal block at the top and the 9-pin header at the bottom. The **right terminal is IN+ (+)** and the **left terminal is IN- (-)**. Identify the terminals using this orientation before wiring.

REQUIRED: COMMON GROUND FOR VOLTAGE MEASUREMENT
Connect **measured-circuit GND, INA236 GND and ESP32 GND together**, even when powered from separate supplies. In the high-side connection described here, source/load negative is the measured-circuit GND.
The INA236 measures bus voltage between **IN- and GND**. Without a shared ground, the voltage reading will be incorrect or undefined. **IN- does not replace the GND connection**: in this setup, IN- connects to load positive.

02 / Ranges and calibration

The sketch sets ADCRANGE = 0 (±81.92 mV). Current is derived from I = Vshunt / R. The bus range is 0 to 48 V DC; bus voltage is measured from IN- to GND [1].

Path	Equivalent R	Use limit	Full scale[1]
A / Header	25 mΩ	3.20 A	≈ 3.277 A
B / Terminal block	31 mΩ	2.58 A[2]	≈ 2.643 A

Calculation for input B

At 3.2 A, input A produces 3.2 × 0.025 = 0.080 V (80 mV). Keeping the same drop for B:
I_B = 0.080 / 0.031 = 2.58065 A → use up to 2.58 A.
The theoretical ADC limit is 0.08192 / 0.031 = 2.64258 A.

[1] Approximate values at nominal ADC range end; the largest positive code is 81.9175 mV. Do not use full scale as an operating margin.
[2] Calculated using the same 80 mV drop, not a thermal qualification of the board. Continuous capability also depends on the shunt, traces, connectors and temperature; module thermal characterization is not available.

Select resistance in the program

Edit **SHUNT_RESISTANCE_OHM** before compiling [2]:

Input	Sketch value
A / 25 mΩ	0.025f
B / 31 mΩ nominal	0.031f
B / current bench adjustment	0.031107383f

The reviewed local copy uses B = 31.107383 mΩ, a one-point adjustment against a 298 mA reference. This is not an independent resistance measurement or a guarantee of accuracy across the range. At this value, 80 mV corresponds to 2.57173 A: use up to **2.57 A**; full scale is ≈ 2.63346 A.

Using 25 mΩ for an actual 31 mΩ path overstates current by about 24%; using 31 mΩ for the 25 mΩ path understates it by about 19.35%. The program recalculates calibration and alert threshold; the constant applies to all detected sensors.

03 / Startup and support

1. Download the sketch from repository [2]. In Arduino IDE, install ESP32 board support and select the appropriate ESP32 board and port. The sketch uses Arduino.h and Wire.h provided by that environment.

2. Open sketch_INA236.ino and select the A or B resistance from the previous page. Check the wiring on page 2. For ESP32, connect Vpu and the ALERT pull-up to 3.3 V, even if the module is powered at 5 V.

3. With supplies off, select one address using ADDR/J1 (0x40, 0x41, 0x42 or 0x43). Use 0x40 to test the default alert. Do not bridge several options at once; verify the detected address at startup.

4. Upload the program, open Serial Monitor at 115200 baud and reset. The sketch checks the four addresses at 100 kHz; confirm OK and System initialized.

5. Apply a small, steady DC load. Check positive current, bus voltage and agreement with a reference instrument. Increase gradually without exceeding the selected input limit.

Alert and readings

The default alert is for positive current above 0.500 A, on 0x40 only. ALERT is open-drain and active LOW; GPIO27 needs the specified pull-up. It is an indication signal and does not disconnect the load. Readings print approximately every 200 ms. Current and Calculated current use the same ADC; agreement does not independently establish accuracy.

Troubleshooting

NOT FOUND / I2C ERROR: check power, common ground, SDA/SCL, Vpu and address selector; reset after correcting.
Current error: check A/B path and configured resistance; verify zero and several steady currents.
Negative current: check IN+ → IN- direction.
Math overflow = YES or clipped reading: reduce current and check configuration. NO does not rule out ADC saturation.

References and scope

[1] Texas Instruments. INA236, SBOSA81D, Rev. D; ranges, registers and interface.
https://www.ti.com/lit/ds/symlink/ina236.pdf
[2] ErickParraFlores. INA236_Sketch; code for ESP32 / Arduino IDE.
https://github.com/ErickParraFlores/INA236_Sketch

Software basis: local sketch_INA236.ino and calibration notes; revision 81f93aa99316, reviewed October 16, 2026. The repository may change. Module pinout and limits were supplied by the designer; B calculations are documented here. The image is a visual reference, not a mechanical drawing.

04 / I2C address selection

The **ADDR / J1** selector is on the right side of the module. It has four rows of two pins. Fit **one jumper across the two pins in the same row** for the required address. Disconnect power from the module and measured circuit before moving the jumper.

Orientation: front view as in Figure 1, blue terminal block at the top and 9-pin header at the bottom. Rows run from **43 at the top to 40 at the bottom**. The silkscreen numbers 40 to 43 indicate hexadecimal addresses, not decimal values.

ADDR / J1 row	I2C address (7-bit)	INA236A A0
43 / top	0x43	SCL
42	0x42	SDA
41	0x41	VS
40 / bottom	0x40	GND

First startup: fit the jumper on row **40** to select **0x40**. Power the system and confirm **Address 0x40: OK** in Serial Monitor at 115200 baud. After changing the address, reset the ESP32 to repeat detection.

Multiple modules: assign a different address to each module. The sketch checks 0x40, 0x41, 0x42 and 0x43, supporting detection of up to four modules on one bus. Share SDA, SCL and GND and set Vpu to the microcontroller logic voltage. Do not connect two modules with the same address on the same bus.

ALERT in the example: enabled only for **0x40** by default. To use the alert at another address, change **ALERT_SENSOR_ADDRESS** in the sketch and connect that module ALERT output to GPIO27. The jumper selects the address; it does not change calibration resistance or select input A/B.

Avoid errors: do not fit multiple jumpers, bridge different rows or fit the jumper vertically. Use one defined position; do not assume removing the jumper selects 0x40.

Technical reference: Texas Instruments, INA236, Table 7-1. A0 mappings apply to the INA236A used by the example. Physical selector orientation follows Figure 1.